

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE:CSA5115 COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO4: *Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO (BL3, BL6)*

Assessment Tool Weightage: 10%

QUESTIONS: 5

TOTAL MARKS: 25

Annexure A

Comparative Analysis

Code of Conduct:

I Nissanki Sri Venkata Lakshmi kavya (192373036)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

N.S.V.L.Kavya

Annexure A

Comparative Analysis

2. Compare transport mode and tunnel mode in IPSec architecture. Evaluate their effectiveness in securing host-to-host versus network-to-network communication.

Transport Mode vs Tunnel Mode in IPSec

Introduction

IPSec (Internet Protocol Security) is a set of protocols used to secure communication over IP networks. It provides **confidentiality, integrity, authentication, and protection against replay attacks**. IPSec mainly operates in two modes: **Transport Mode** and **Tunnel Mode**. The choice between them depends on whether communication is between individual hosts or between entire networks.

1. Transport Mode

In **Transport Mode**, IPSec protects the payload of the original IP packet while the original IP header remains unchanged.

Structure:

Original IP Header | IPSec Header | Encrypted/Protected Data

Features

- Protects the IP payload.
- The original source and destination IP addresses remain visible.
- Has less overhead because a new IP header is not added.
- Generally used for **host-to-host communication**.
- Suitable when both communicating devices support IPSec directly.

Example

A company has two servers that need to communicate securely:

Server A → IPSec Transport Mode → Server B

The data between the two servers is protected, while their IP addresses remain visible.

2. Tunnel Mode

In **Tunnel Mode**, the entire original IP packet, including its original IP header, is protected and encapsulated inside a new IP packet.

Structure:

New IP Header | IPSec Header | Encrypted Original IP Packet

Features

- Protects the entire original IP packet.
- Hides the original source and destination addresses from outside observers.
- Adds a new IP header for routing.
- Provides greater protection and is commonly used for **network-to-network communication**.
- Commonly used in VPNs.

Example

Suppose a company has two offices:

Head Office Network → VPN Gateway → Internet → VPN Gateway → Branch Network

Tunnel Mode can encrypt and encapsulate the original packets travelling between the two networks.

3. Comparison

Feature	Transport Mode	Tunnel Mode
Protected data	IP payload	Entire original IP packet
Original IP header	Remains visible	Encapsulated/protected
New IP header	No	Yes
Overhead	Lower	Higher
Main use	Host-to-host	Network-to-network
Common application	Secure server communication	VPN
Address hiding	Limited	Better
Performance	Generally faster	More processing overhead

4. Effectiveness for Different Communication Types

Host-to-Host Communication

Transport Mode is more effective for host-to-host communication because it directly protects the data exchanged between two individual hosts. It has lower overhead and is efficient when both hosts support IPSec.

For example:

Host A ↔ Host B

Transport Mode is suitable when the goal is to securely communicate between two known endpoints.

Network-to-Network Communication

Tunnel Mode is more effective for network-to-network communication. It allows entire IP packets from one private network to be protected while travelling through an untrusted network such as the Internet.

For example:

Office A Network ↔ VPN Gateway ↔ Internet ↔ VPN Gateway ↔ Office B Network

Tunnel Mode is therefore commonly used to create **site-to-site VPNs**.

Conclusion

Both IPSec modes provide strong security, but they serve different purposes. **Transport Mode** is best suited for **host-to-host communication** because it protects the payload with less overhead. **Tunnel Mode** is better for **network-to-network communication** because it protects and encapsulates the complete original IP packet and is well suited for VPNs. Therefore, organizations should select the mode according to their communication architecture and security requirements.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand